

Transactions

A database transaction groups related changes into a unit that commits or rolls back. For a document workflow, authoritative metadata can record which processing stage completed. A SQL transaction does not automatically roll back writes made to an external vector database or object store. In this course transactions are a prerequisite for idempotency, and idempotency is a prerequisite for safe retries.

Idempotency

Idempotency means repeating an operation has the same intended effect as performing it once. Deterministic chunk identifiers and upserts can prevent a repeated ingestion attempt from creating duplicate vectors. Transactions are a prerequisite for studying idempotency in this workflow, and idempotency is a prerequisite for safe retries. Idempotency does not mean every attempt succeeds; the operation must still report failures.

Safe Retries

Safe retries repeat a transiently failed operation under a bounded attempt budget. Their direct prerequisite is idempotency, whose prerequisite is transactions. Use a delay or backoff when a provider is busy and stop retrying permanent input errors. If an earlier write may already have succeeded, deterministic identifiers help make the next attempt safe. Blindly repeating a non-idempotent action can duplicate data or effects.

Compensating Cleanup

Compensating cleanup removes partial derived artifacts after a workflow fails across services. A PostgreSQL rollback cannot erase points already written to Qdrant. Delete by the exact upload identifier and preserve authoritative metadata until cleanup can be retried. Cleanup is not a distributed atomic transaction: it can also fail, so record the failure and keep operations idempotent.

Vector Search

Vector search compares numeric embeddings to locate passages with similar meaning. Filter by the allowed upload identifiers before returning results. A similarity score does not prove that a passage supports the answer. Reranking can compare the question and each candidate more carefully; an evidence threshold can reject weak matches. Vector search complements a concept graph, which stores explicit typed relationships such as prerequisites.

Readiness

Readiness decides whether a service can accept useful traffic. Required dependencies such as the authoritative database must work. An optional concept graph may be reported as degraded while vector-based answers continue. A healthy process does not prove that an external AI provider has quota. Readiness and rate-limit failures require different operator responses.